$(n \to n-1)$ (c, prev)

Applications of Recursion :→

Back Tracking

{N Queens} (4 Queens)

Optimal 4x4 N-Queen Solution

1 Queen in each row

* We try to place a queen in the 1st row 1st col.
* While placing if it violates rule, we backtrack.
* We try in the next col & so on.
* The backtracking is done using Recursion.

Traversals in Trees & Graphs

(In | Pre | Post)

(Recursion)

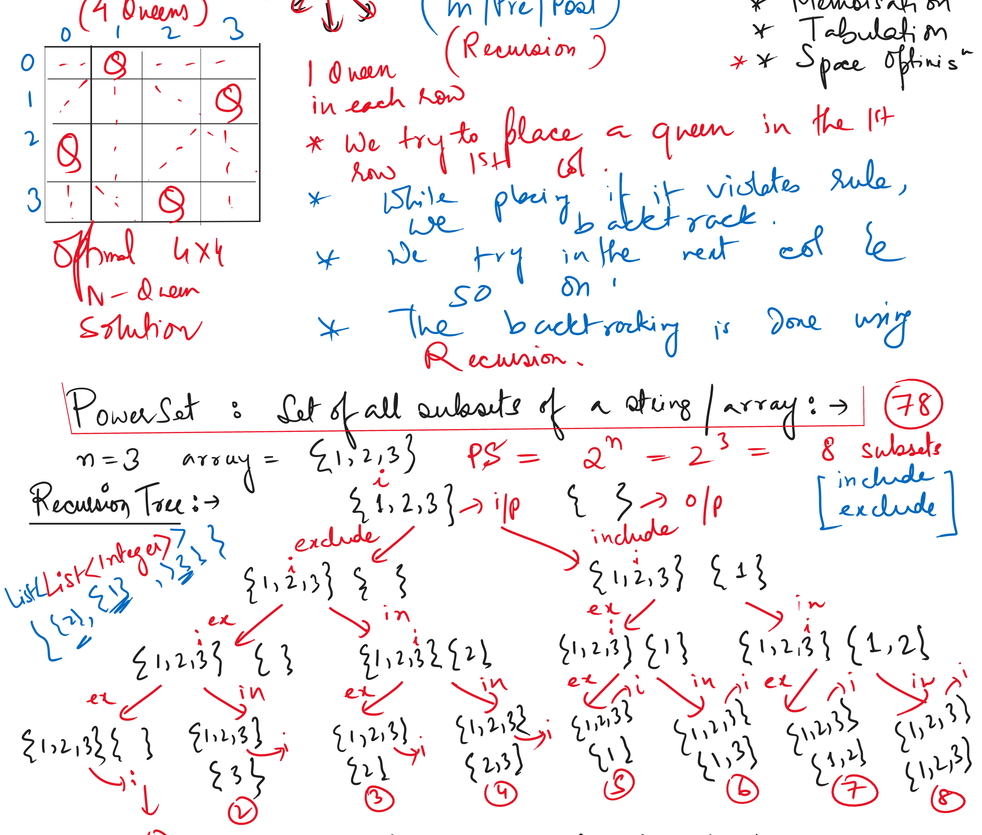
Dynamic Programming

* Recursion
* Memoisation
* Tabulation
* Space Optimis^n

PowerSet : Set of all subsets of a string / array :→ (78)

$n=3$ array = {1,2,3} PS = $2^n = 2^3 =$ 8 subsets [include exclude]

Recursion Tree :→



The moment i goes beyond the length of the array, we get one of our outputs. So, to get the next output we need to go back to the beginning → Backtrack

ans { { }, {1}, {2} and so on }

output = {2} [Latest] {1,2,3} { }

ans.add(new ArrayList<>(output));

output.remove(output.size()-1); {1,2,3} {1}

So we remove the last output which was stored in the answer [Backtracking] [Keeping the o/p fresh for latest answer]

Important Backtracking Interview Coding Questions :→

× ① Rat In A Maze

× ② Sudoku Solver

③ Subsets of a String / Array (LC-78) { } "x" {1,2,3} (8) "abc" (7)

8 to 12 LPA ④ Phone Keypad Problem

(Letter Combinations of a Phone Number) (LC-17)